**Ans: A**

**24** Two point charges each of charge $Q$ at a separation of $r$ repel each other with a force $F$. Each charge is increased by adding $3Q$ and the separation is increased to $2r$. What is the force between the charges?

A $F$ B $9F/4$ C $4F$ D $8F$

**Ans: C**

Original force:

$$F = \frac{kQ \times Q}{r^2} = \frac{kQ^2}{r^2}$$

New force:

$$F_{new} = \frac{k \times 4Q \times 4Q}{(2r)^2} = \frac{16kQ^2}{4r^2} = \frac{4kQ^2}{r^2}$$

Ratio:

$$\frac{F_{new}}{F} = \frac{4kQ^2/r^2}{kQ^2/r^2} = 4 \Rightarrow F_{new} = 4F$$